# SAMIULLAH PANWAR

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## EDUCATION

**Mehran University of Engineering and Technology** Jamshoro, Pakistan
Bachelor of Engineering in Electrical Engineering Dec 2026
CGPA: 3.8/4.0
Class Rank: 2 of 92
Thesis: Automatic Hybrid Solar Cleaning Robot (In Progress)

## RESEARCH INTERESTS

Renewable Energy Systems, Grid Integration, Power System, Smart Grids, Microgrids, AI Data centres, EV & BESS.

## RESEARCH EXPERIENCE

**Department of Electrical Engineering, Mehran University of Engineering and Technology** Jamshoro, Pakistan
Undergraduate research student Oct 2025 – Present

- Designing a hybrid air-and-water solar panel cleaning robot with an Esp32-based control system, rail-mounted rack-and-pinion movement, and dual motor drivers to reduce dust-related efficiency losses in arid climates.
- Leading component selection, market survey, and procurement for the prototype; conducting techno-economic analysis to evaluate cost-effectiveness against efficiency gains.

**Department of Electrical Engineering, Mehran University of Engineering and Technology** Jamshoro, Pakistan
Undergraduate research student Dec 2025 – April 2026

- Derived and simulated the swing equation in MATLAB for a 2MW wind generator to analyze rotor angle stability under fault conditions, determining a Critical Clearing Time of 0.286s.
- Trained a Random Forest ML model on 3,456 simulation samples to predict generator stability from fault duration and system parameters, achieving 99.1% test accuracy.

## PRESENTATIONS

**Institute of Mechanical Engineers, IMechE-MUET chapter** (Speak Out for Engineers, SOfe Locals) Sep 2026
Blackouts and Bytes: Can Our Energy Grid Survive the AI Era

## VOLUNTEER EXPERIENCE

**Inclusion and Empowerment Organization** 2025 – 2026
*Volunteer — Hyderabad, Pakistan*

- Restored daily mobility for dozens of families, equipped participants with functional digital literacy for remote opportunities, and learned how community driven advocacy creates sustainable.
- Led community outreach to distribute wheelchairs.

**Department of Electrical Engineering, MUET** Oct 2024
*Organizer — Jamshoro, Pakistan*

- Organized department wide multi sport event (50+ participants, 7 sports) as part of an 8-member team; managed budget and registrations.
- Handled budget and registration for a multi-sport tournament with 50+ participants across 7 sports, as part of an 8-person organizing team

## PROFESSIONAL EXPERIENCE

**Vivo Pakistan** July 2026

*Maintenance Engineer Intern — Hyderabad, Pakistan*

- Diagnosed and repaired 15+ complex hardware and component-level faults daily across mobile devices, achieving a 98% repair success rate and restoring full functionality under tight turnaround windows.
- Isolated circuit and board-level issues using precision diagnostic tools, reducing average service turnaround time by 20% and improving daily customer throughput.
- Streamlined testing workflows to accurately identify micro-hardware defects, accelerating fault resolution times and contributing to a 15% increase in post-repair customer satisfaction ratings.

**Islamabad Electric Supply Company** June 2026

*Intern — Islamabad, Pakistan*

- Supported operations across 3, 132 kV and 66 kV grid stations, assisting with load balancing and fault diagnostics on a distribution ring network serving over 3 million consumers.
- Conducted preventive maintenance and real-time power flow monitoring on 15 transformers and feeders, helping sustain 99.9% uptime and uninterrupted supply.
- Developed practical understanding of large-scale utility infrastructure and SCADA-based monitoring systems, supporting faster fault detection and response.

**Master Wind Energy Limited** May 2026

*Intern — Jhimpir, Pakistan*

- Optimized operational reliability at a 52.8 MW wind plant by systematically monitoring daily performance metrics and diagnosing critical equipment failures, including gearbox faults and derating issues, across 33 GE turbines using industry-standard GE SCADA software.
- Performed annual maintenance procedures and safety drills, including lock-out tag-out.
- Visited Sachal Wind Energy Limited (50 MW) to benchmark Goldwind turbine technology against GE systems.

**National Grid Company** Jul 2025 – Aug 2025

*Intern — Hyderabad, Pakistan*

- Analysed protection schemes across 220/132/11 kV systems, including relays, circuit breakers, and transformers, and independently modelled the local grid network in ETAP, running load flow analyses to assess system performance and stability.
- Investigated distribution-side operations spanning feeders, transformers, and metering/billing systems serving local households, and conducted site visits to 132 kV/220 kV transmission towers and substations to map the end to-end power delivery chain from generation to consumer.

**Gharo Solar Limited** May 2025 – Jul 2025

*Intern — Gharo, Pakistan*

- Diagnosed and resolved faults (line-to-line short circuits) and solar tracker malfunctions (motor, gear, shaft issues) at a 50 MW solar PV plant, progressing from guided to independent troubleshooting; supported daily control room and shift operations.
- Performed equipment testing (transformers HV/LV, DRM), CT/PT, and GIS SF6 breaker dew point tests; studied PLCs, protection relays, wave traps, weather monitoring station (WMS) instrumentation, and tracking systems.

**Jamshoro Power Company Limited** Jan 2024

*Intern — Jamshoro, Pakistan*

- Gained foundational exposure to thermal power generation at an 880 MW, 4-unit thermal power station, observing electrical systems of boilers, turbines, and generators.

## LEADERSHIP EXPERIENCE

**IEEE MTT-S MUET Chapter** Jan 2025 – Present

*Secretary — Jamshoro, Pakistan*

- Co-led a core team of 20 to organize technical and community events for a 50+ member chapter, reaching 50–100 attendees per event; collaborated with IEEE WIE, SIGHT, Comsoc and PES on joint events.
- Organized RF and antenna technology sessions featuring international speakers from Italy.
- Managed event logistics, including speaker coordination, budgeting, and online platform setup, over a 2+ year term as Secretary.

**Department of Electrical Engineering, MUET** Jul 2024 – Nov 2024

*Class Representative — Jamshoro, Pakistan*

- Elected as academic liaison (Class Representative) for a class of 44 Electrical Engineering students, serving as the primary point of contact between faculty and students for one semester, relaying announcements, schedule changes, and academic concerns, while organizing extracurricular activities to strengthen class engagement.
- Coordinated directly with course instructors to resolve timetable conflicts, exam scheduling, and classroom issues on behalf of the class, ensuring smooth academic operations throughout the semester.

## SKILLS AND COMPETENCES

**Technical Skills:** ETAP, MATLAB/Simulink, Multisim, Arduino, AVR Studio, Proteus, PLC Ladder logic C++ & Python (basics), AI, Microsoft Applications (Excel, Word, and PowerPoint)

**Personal Skills:** Problem-solving, Teamwork, Leadership, Creativity, Adaptability, Time management

## PROFESSIONAL BODY

Student Member: Institute of Electrical and Electronics Engineer (IEEE) 2023-2026
Student Member: IEEE Societies (MTT-S, PES, WEI, ComSoc, YP, CAS, EMBS, IAS, Consultant Network

## SCHOLARSHIPS, HONORS & AWARDS

- Recipient, Michael Taiwo Annual Scholarship [69 / 9,198] 2026
- Recipient, Prime Minister's Youth Laptop Scheme, Government of Pakistan 2026
- Recipient, Sindh Education Endowment Fund (SEEF) Scholarship, Government of Sindh 2024-2026
- Recipient, Merit List Scholarship, Department of Electrical Engineering, MUET 2024-2026

## LANGUAGES

Urdu & Sindhi (Native), English (B2 – Upper Intermediate), German (A1 – Beginner)

## CERTIFICATIONS

- PLC Programming — LinkedIn Learning
- AutoCAD — LinkedIn Learning
- MATLAB — LinkedIn Learning
- Project Management — LinkedIn Learning
- Python Programming — Coursera
- AI Prompting / Prompt Engineering — Coursera